Support Vector Machine (SVM)

Introduction

- Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression tasks. It is also called as Maximum Margin Classifier
- SVM finds a hyperplane that segregates the labeled dataset into two classes.
- In machine learning and higher-dimensional geometry, **a hyperplane is a flat, n-1 dimensional subspace that divides an n-dimensional space into two regions**, often used as a decision boundary in classification or regression tasks.
- It is particularly effective for high-dimensional spaces and is widely used in machine learning applications such as text classification, image recognition, and bioinformatics.

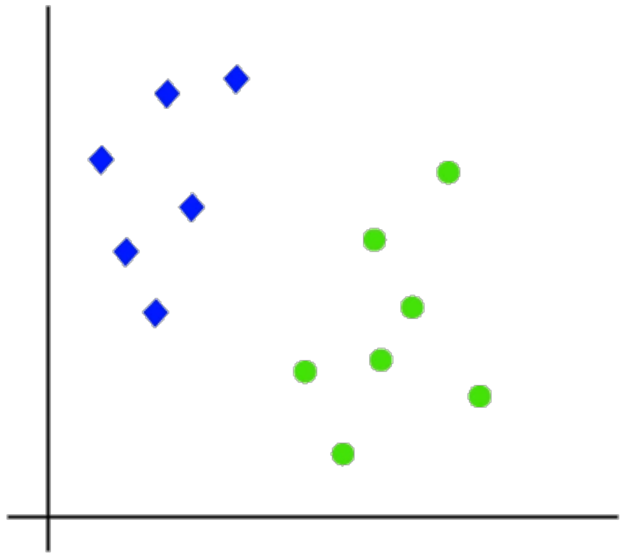
Types of SVM

Support Vector Machines (SVMs) come in two main types

- **Linear SVM**
  - Used for linearly separable data
- **Non-linear SVM**
  - Which handles complex data using kernel functions like RBF and polynomial.

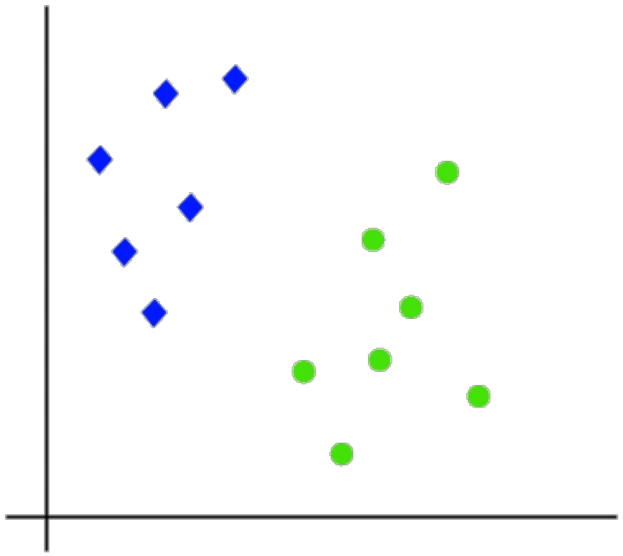
Linear SVM

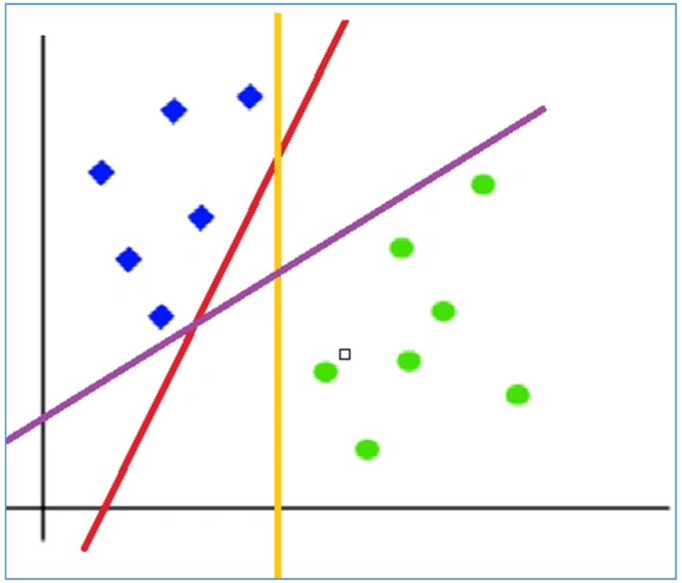
- When the data is linearly separable only then we can use Linear SVM.
- Linearly separable means that the data points can be classified into 2 classes by using a single straight line(if 2D).



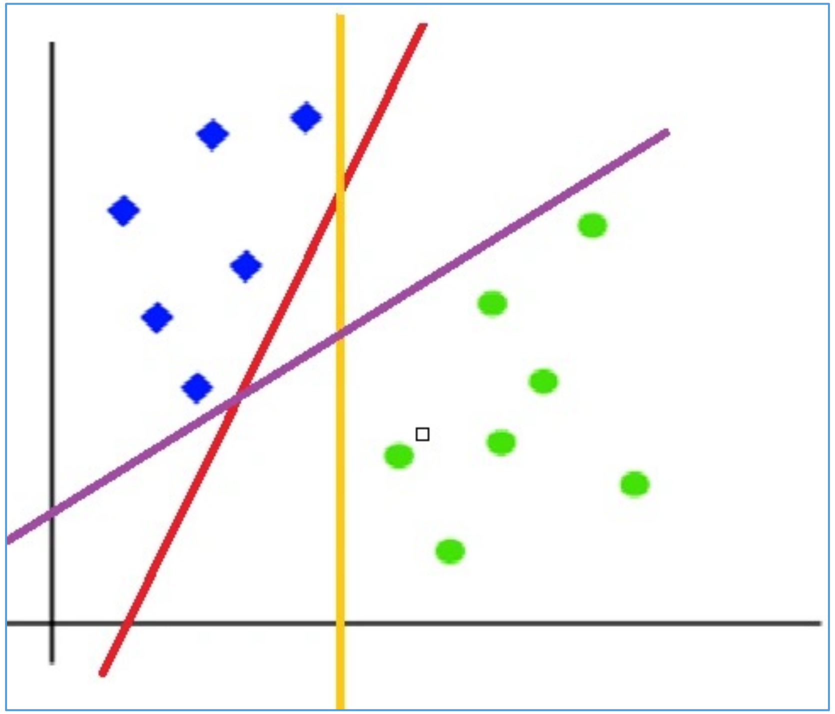
Hyperplane

- A hyperplane is a **decision boundary** that separates data points into different classes in a high-dimensional space.
- **In two-dimensional space, a hyperplane is simply a line** that separates the data points into two classes
- When we add the new testing data, whatever side of the hyperplane it goes will eventually decide the class that we assign to it





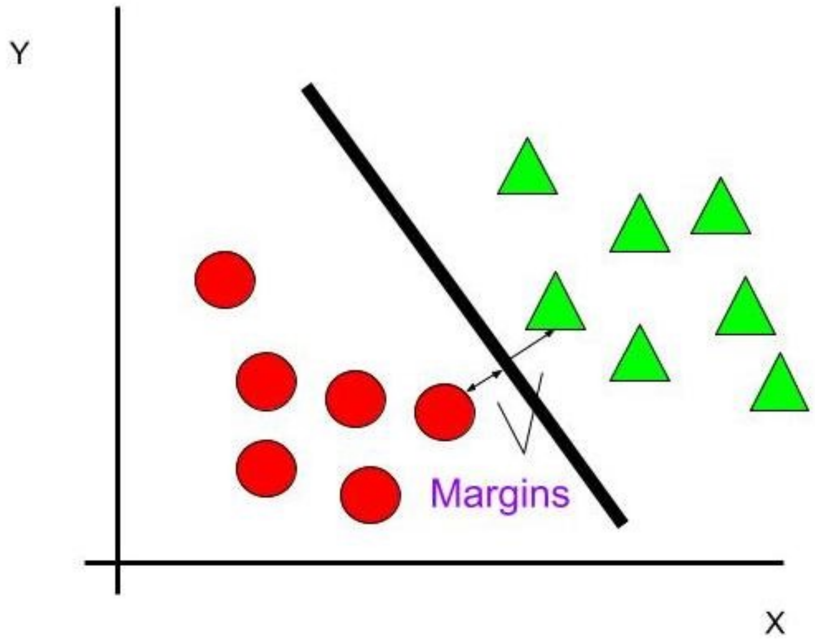
Hyperplane



To segregate the dataset into classes we need the hyperplane. **To choose the right hyperplane we need margin**. Margin is the distance between the hyperplane and the closest point from either set.

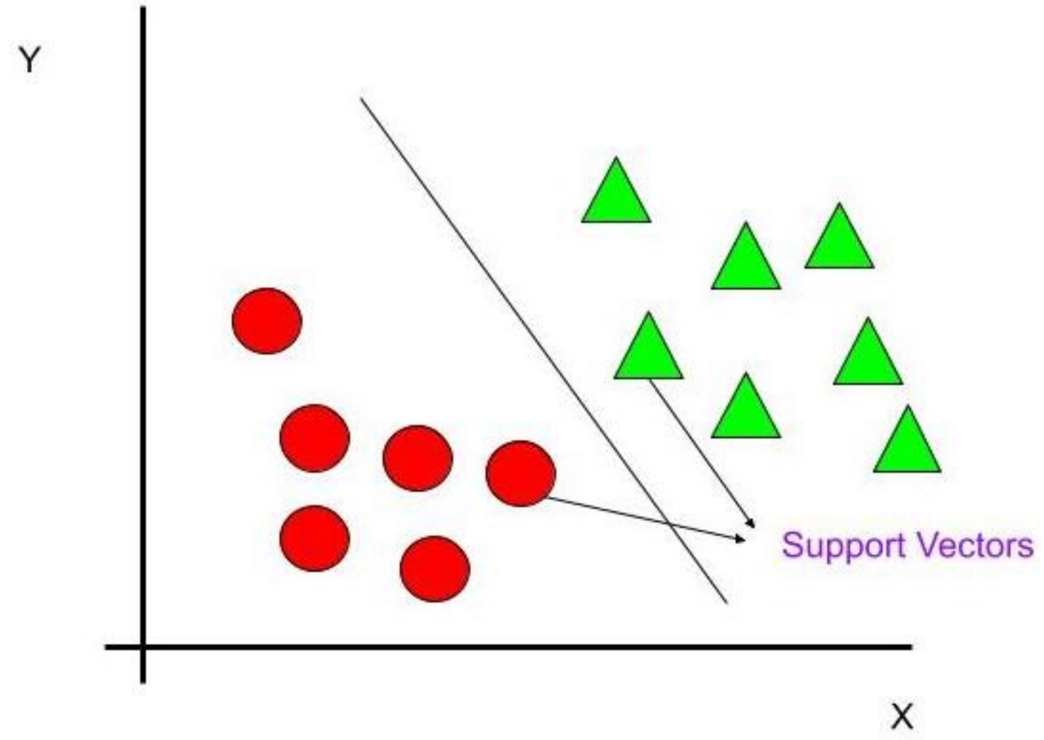
Margin

The margin is the distance between the hyperplane and the closest data points from either class. SVM aims to maximize this margin to ensure better generalization.



Support Vectors

Support vectors are the data points closest to the hyperplane. These points play a crucial role in defining the position and orientation of the decision boundary.



- SVMs are commonly used within classification problems.
- They distinguish between two classes by finding the optimal hyperplane that maximizes the margin between the closest data points of opposite classes.
- The number of features in the input data determine if the hyperplane is a line in a 2-D space or a plane in a n-dimensional space.
- Since multiple hyperplanes can be found to differentiate classes, maximizing the margin between points enables the algorithm to find the best decision boundary between classes.
- This, in turn, enables it to generalize well to new data and make accurate classification predictions.
- The lines that are adjacent to the optimal hyperplane are known as support vectors as these vectors run through the data points that determine the maximal margin

Support Vector Machine (SVM) is a supervised learning algorithm used for classification and regression tasks. It works by finding the best **decision boundary** (hyperplane) that separates data points of different classes with the **maximum margin**.

Finding the Optimal Hyperplane

We are given a training dataset of n points of the form

$$(x_1,y_1),\ldots\ldots,(x_n,y_n)$$

where,

- $\mathbf{y_i}$ are either 1 or -1, each indicating the class to which the point x_i belongs
- Each $\mathbf{x_i}$ is a p-dimensional vector
- We want **to find the "maximum-margin hyperplane"** that divides the **group of points x_i for which y_i =1** from the **group of points for which y_i = -1**, which is defined so that the distance between the hyperplane and the nearest point x_i from either group is maximized

Any hyperplane can be written as the set of points satisfying

$$\mathbf{w}^{\mathrm{T}}\mathbf{x} + b = 0$$

The hyperplane is defined by the equation:

$$w \cdot x + b = 0$$

where:

- w is the weight vector (normal to the hyperplane),
- x is the input feature vector,
- b is the bias term.

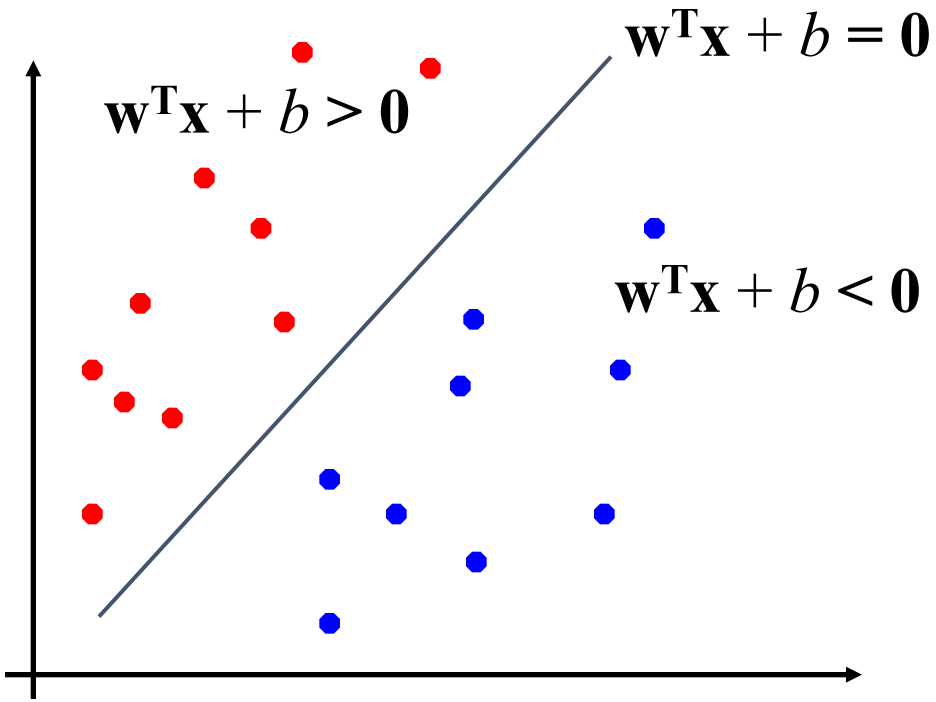
For a binary classification problem with labels $y \in \{+1, -1\}$, the decision rule is:

$$y = \text{sign}(w \cdot x + b)$$

- Our Goal is to find out w and b such that

$$\mathbf{w}^{\mathrm{T}}\mathbf{x} + b > 0 \qquad \text{for } \mathbf{y_i} = +1$$

$$\mathbf{w}^{\mathrm{T}}\mathbf{x} + b < 0 \qquad \text{for } \mathbf{y_i} = -1$$



$$f(\mathbf{x}) = \mathrm{sign}(\mathbf{w}^{\mathrm{T}}\mathbf{x} + b)$$

Example

Feature Vector x

- Suppose each data point has two features (Height, Weight)

x=(170,65)

Weight Vector w

- Weights determine **orientation of the hyperplane** w=(w1,w2)

w=(2,3)

Dot Product $\boldsymbol{w^T x}$

$$w^T x = w_1 x_1 + w_2 x_2$$

$$w^T x = 2x_1 + 3x_2$$

Bias b

- Bias shifts the line **left/right or up/down**.

b=−6

Full Equation of Hyperplane

$$2x_1 + 3x_2 - 6 = 0$$

- This is the **decision boundary**.
- All points satisfying this equation lie **on the hyperplane**.

Condition	Meaning
$w^T x + b = 0$	point lies **on hyperplane**
$w^T x + b > 0$	one class
$w^T x + b < 0$	other class

So SVM classification rule is

$$f(x) = sign(w^T x + b)$$

$$2x_1 + 3x_2 - 6 = 0$$

Example 1

Point:

$$x = (1, 1)$$

Substitute:

$$2(1) + 3(1) - 6$$

$$= 2 + 3 - 6$$

$$= -1$$

Since result < **0**

Point belongs to **Class −1**

Example 2

Point:

$$x = (3, 2)$$

Substitute:

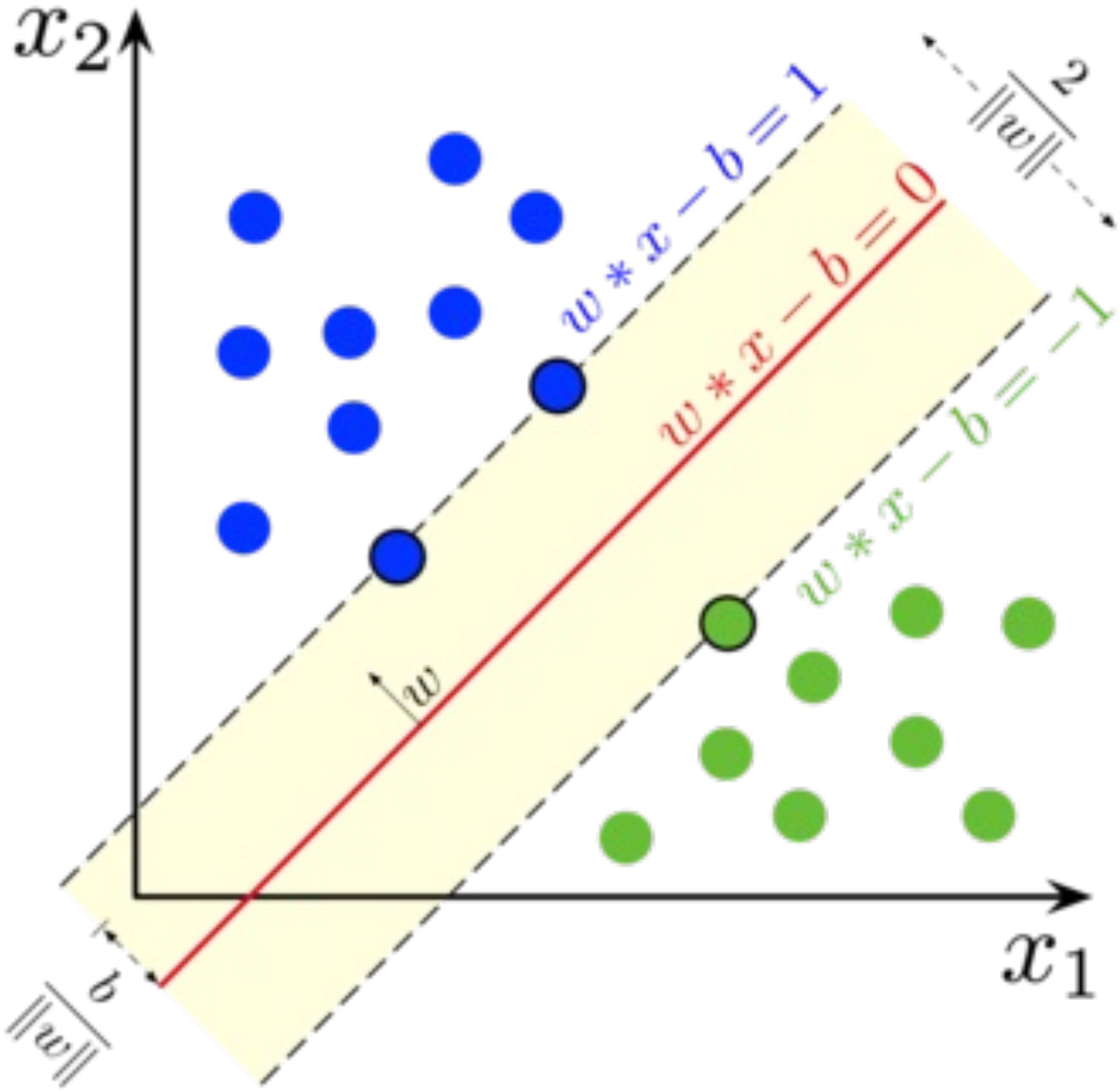
$$2(3) + 3(2) - 6$$

$$= 6 + 6 - 6$$

$$= 6$$

Since >**0**

Point belongs to **Class +1**



x_2
x_1
$w * x - b = 1$
$w * x - b = 0$
$w * x - b = -1$
$\frac{2}{\|w\|}$
$\frac{b}{\|w\|}$
w

Math behind SVM

$$\boxed{\max \frac{2}{\|w\|} \text{ such that } w^T x_i + b \quad \begin{matrix} \geq 1 & \text{if } Y_i = +1 \\ \leq -1 & \text{if } Y_i = -1 \end{matrix}}$$

Maximum Margin Concept

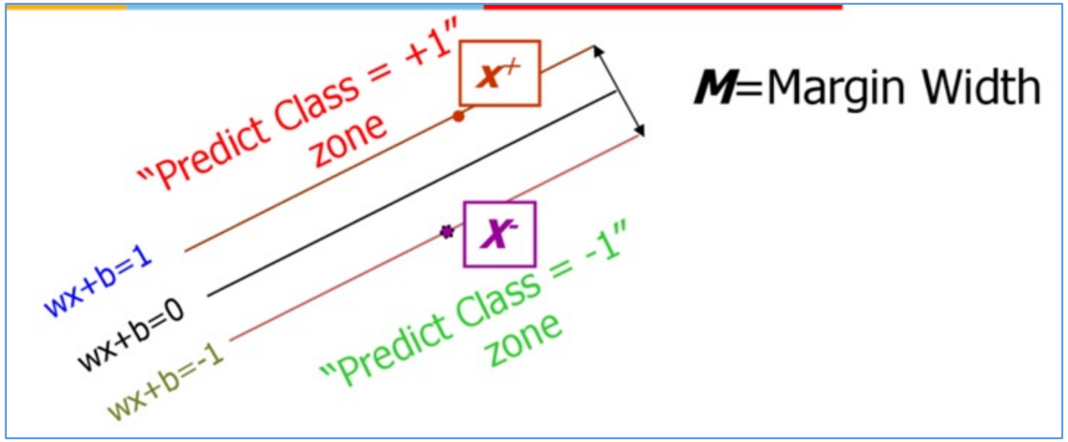
SVM aims to maximize the margin M, which is the distance between the closest data points (support vectors) and the hyperplane. The margin is given by:

$$M = \frac{2}{||w||}$$

Maximizing M is equivalent to minimizing $\frac{1}{2}||w||^2$, subject to the constraint:

$$y_i(w \cdot x_i + b) \geq 1, \quad \forall i$$

This ensures that all points are correctly classified with at least margin M.



Problem Statement:

We have the following training dataset with two features (x_1, x_2) and two class labels (y):

x_1	x_2	y
2	2	$+1$
4	4	$+1$
4	0	-1
0	0	-1

We need to determine the **optimal separating hyperplane** using a **Linear SVM**.

Step 1: Define the SVM Optimization Problem

For a **hard-margin SVM**, we solve:

$$\min_{\mathbf{w},b} \frac{1}{2}||\mathbf{w}||^2$$

subject to the **constraint**:

$$y_i(\mathbf{w} \cdot \mathbf{x}_i + b) \geq 1, \quad \forall i$$

where:

- $\mathbf{w} = [w_1, w_2]$ is the normal vector to the hyperplane.
- b is the bias term.
- y_i is the class label.

Step 2: Solve for $\mathbf{w}$ and b

The decision boundary is:

$$w_1 x_1 + w_2 x_2 + b = 0$$

Finding Support Vectors

Support vectors are the points that satisfy:

$$y_i(\mathbf{w} \cdot \mathbf{x}_i + b) = 1$$

From our dataset, we identify two support vectors:

- $[2, 2]$ (Class +1)
- $[4, 0]$ (Class -1)

Using these, we set up two equations:

$$w_1(2) + w_2(2) + b = 1$$

$$w_1(4) + w_2(0) + b = -1$$

From

$$4w_1 + b = -1$$

$$b = -1 - 4w_1$$

Substitute into the first equation

$$2w_1 + 2w_2 + (-1 - 4w_1) = 1$$

$$-2w_1 + 2w_2 - 1 = 1$$

$$-2w_1 + 2w_2 = 2$$

$$-w_1 + w_2 = 1$$

$$w_2 = w_1 + 1$$

Use SVM geometric property

Vector between support vectors:

$$(2,2) - (4,0) = (-2,2)$$

The **normal vector** w is parallel to this vector.

$$w = \lambda(-2,2)$$

So

$$w_1 = -2\lambda, \quad w_2 = 2\lambda$$

Find λ

From

$$w_2 = w_1 + 1$$

Substitute

$$2\lambda = -2\lambda + 1$$

$$4\lambda = 1$$

$$\lambda = \frac{1}{4}$$

Compute w

$$w_1 = -\frac{1}{2}$$

$$w_2 = \frac{1}{2}$$

$$w = \left(-\frac{1}{2}, \frac{1}{2}\right)$$

Compute b

Using

$$4w_1 + b = -1$$

$$4(-\frac{1}{2}) + b = -1$$

$$-2 + b = -1$$

$$b = 1$$

Final Answer

Weight vector

$$w = \left(-\frac{1}{2}, \frac{1}{2}\right)$$

Bias

$$b = 1$$

Decision boundary

$$-\frac{1}{2}x_1 + \frac{1}{2}x_2 + 1 = 0$$

Multiply by 2:

$$-x_1 + x_2 + 2 = 0$$

or

$$x_2 = x_1 - 2$$

Hard Margin SVM

- **Hard margin SVM assumes that the data is perfectly linearly separable.**
- It tries to find a hyperplane that **separates the classes without any misclassification**.
- **Key Idea**
  - No point is allowed inside the margin
  - No misclassification allowed

Soft Margin SVM

- Soft margin SVM **allows some classification errors** so that the model becomes **robust to noise and overlapping classes**.
- Soft margin SVM introduces **slack variables ξ_i** that allow some training samples to lie:
  - Inside the margin
  - On the wrong side of the decision boundary
- This enables the classifier to **balance margin maximization and classification error**.

Soft margin SVM **allows some classification errors** so that the model becomes **robust to noise and overlapping classes**.

It introduces **slack variables** ξ_i.

Constraint

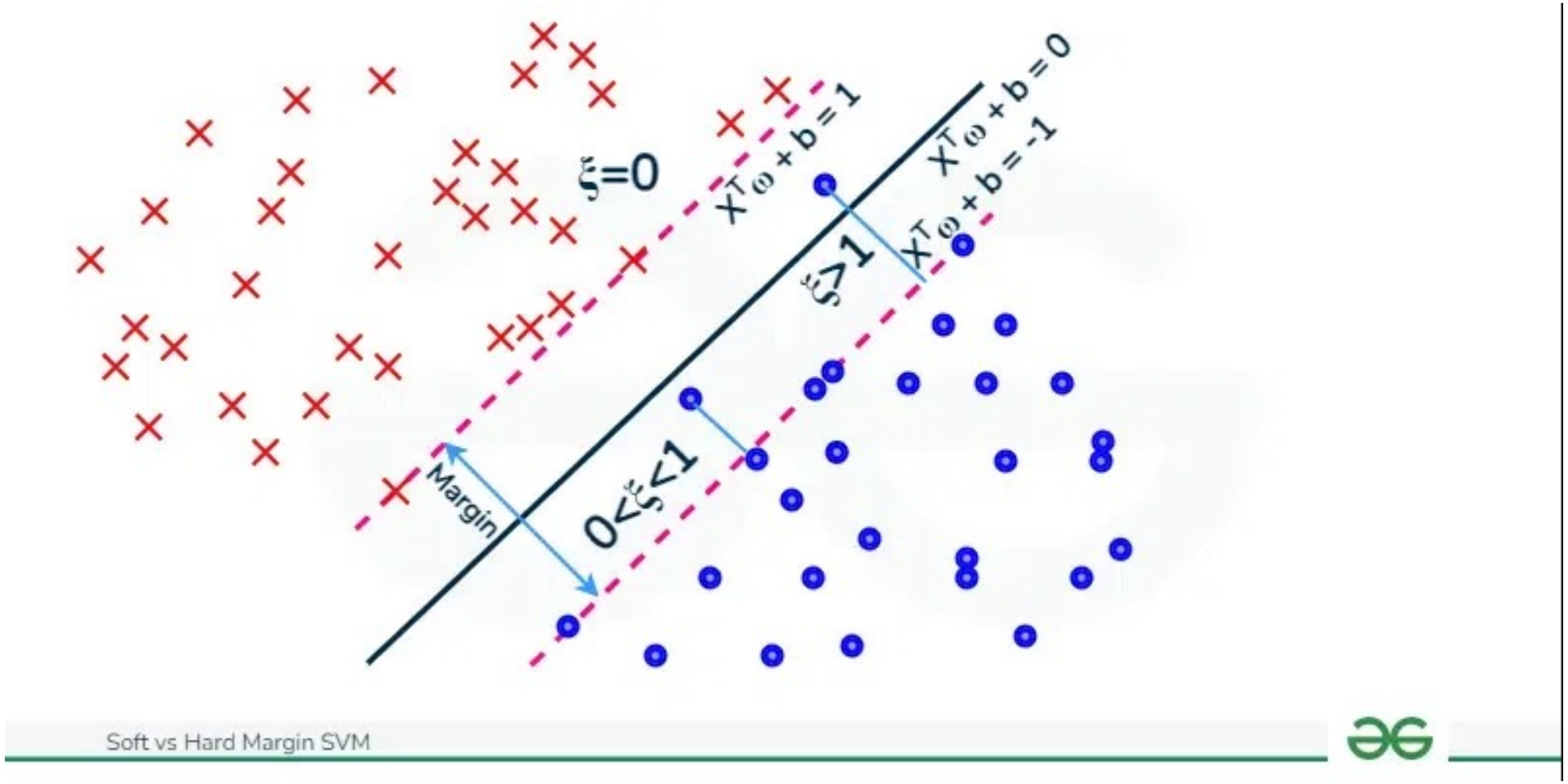
$$y_i(w^T x_i + b) \geq 1 - \xi_i$$

Where

$$\xi_i \geq 0$$

- $\xi_i = 0$ → correctly classified
- $0 < \xi_i < 1$ → inside margin
- $\xi_i > 1$ → misclassified

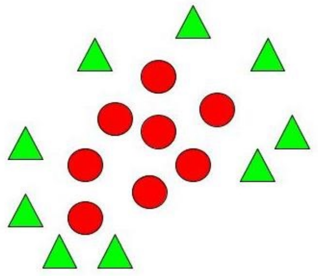
Soft Margin



$X^T\omega + b = 0$
$X^T\omega + b = 1$
$X^T\omega + b = -1$
$\xi=0$
$\xi>1$
$0<\xi<1$
Margin
Soft vs Hard Margin SVM

Non – Linear SVM.

- When we can easily separate data with hyperplane by drawing a straight line is Linear SVM.
- When we cannot separate data with a straight line we use **Non – Linear SVM**. In this, we have **Kernel functions.** They transform non-linear spaces into linear spaces. It transforms data into another dimension so that the data can be classified.
- It transforms two variables x and y into three variables along with z. Therefore, the data have plotted from 2-D space to 3-D space. Now we can easily classify the data by drawing the best hyperplane between them

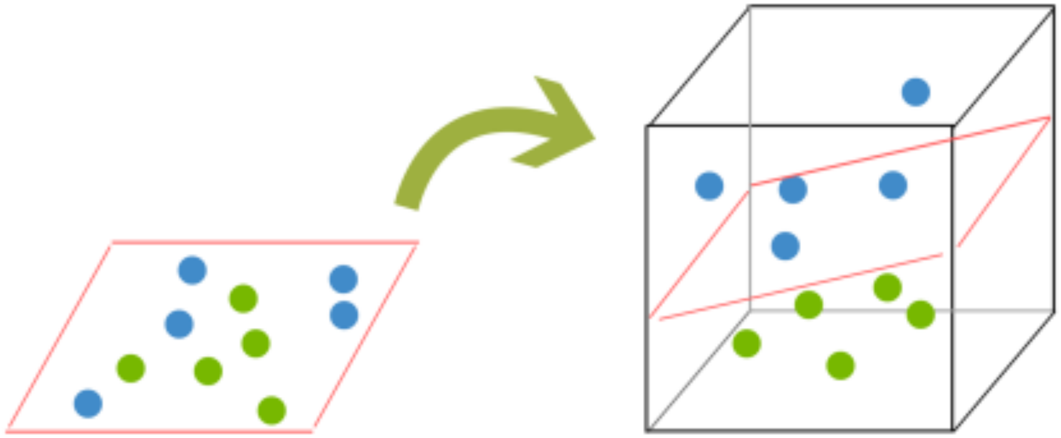


- Much of the data in real-world scenarios are not linearly separable, and that’s where nonlinear SVMs come into play.
- In order to make the data linearly separable, preprocessing methods are applied to the training data to transform it into a higher-dimensional feature space. That said, higher dimensional spaces can create more complexity by increasing the risk of overfitting the data and by becoming computationally taxing.
- The “kernel trick” helps to reduce some of that complexity, making the computation more efficient, and it does this by replacing dot product calculations with an equivalent kernel function.
- There are a number of different kernel types that can be applied to classify data. Some popular kernel functions include:
  - Polynomial kernel
  - Radial basis function kernel (also known as a Gaussian or RBF kernel)
  - Sigmoid kernel

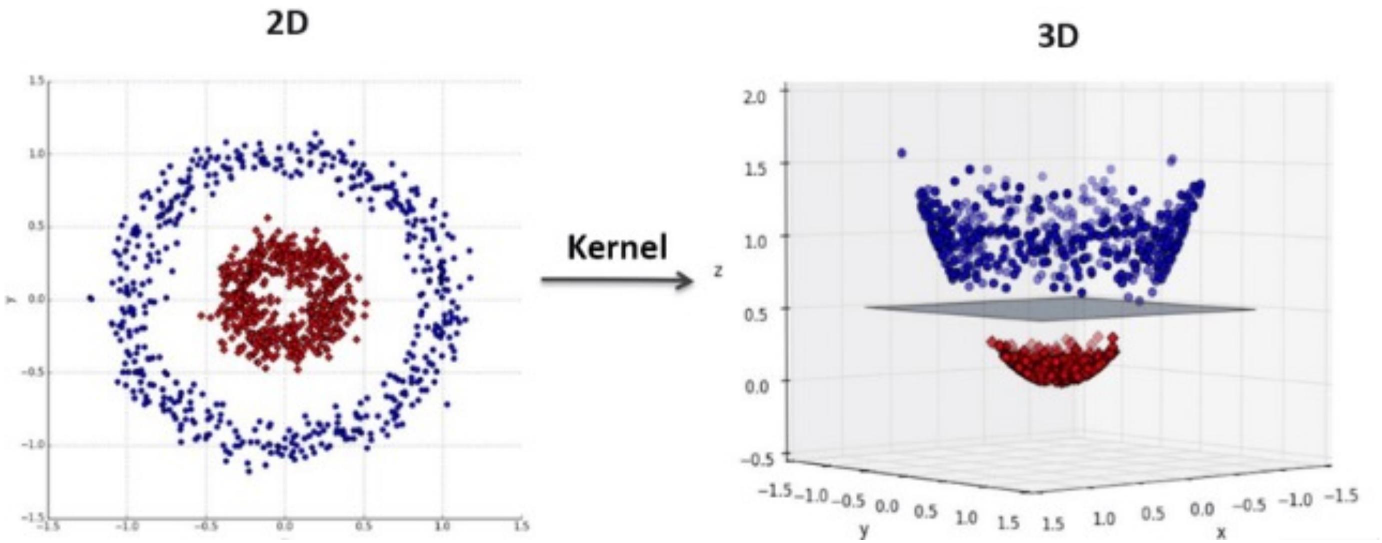
Kernel Trick

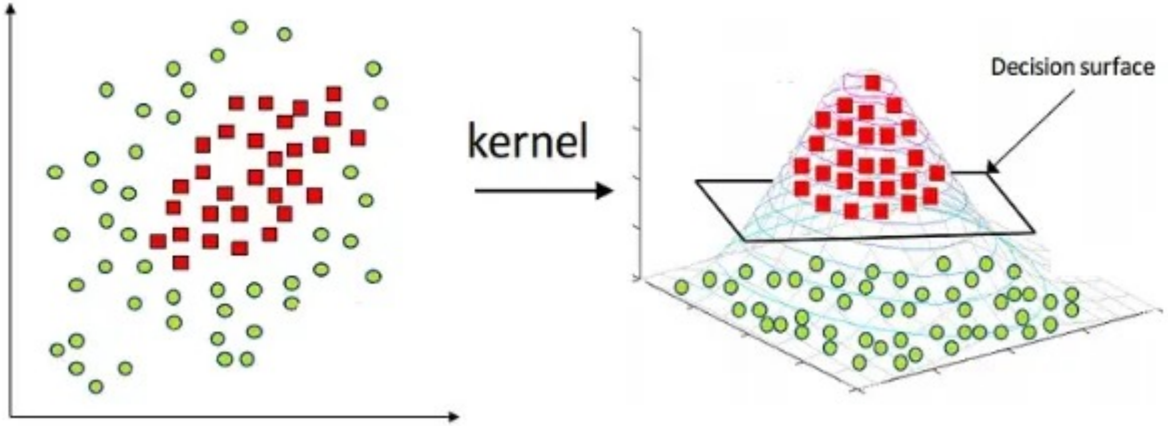
Instead of directly working with the dot product $x_i \cdot x_j$, we use a kernel function $K(x_i, x_j)$ that implicitly maps data into a higher-dimensional space:

- **Linear Kernel:** $K(x_i, x_j) = x_i \cdot x_j$
- **Polynomial Kernel:** $K(x_i, x_j) = (x_i \cdot x_j + c)^d$
- **Radial Basis Function (RBF) Kernel:** $K(x_i, x_j) = e^{-\gamma ||x_i - x_j||^2}$
- **Sigmoid Kernel:** $K(x_i, x_j) = \tanh(\alpha x_i \cdot x_j + c)$



Support Vector Machines (SVM) can handle highly non-linear problems using a kernel trick which implicitly maps the input vectors to higher-dimensional feature spaces.


2D
Kernel
3D
z
y
x


kernel
Decision surface

SVM

- The SVM algorithm is widely used in machine learning as it can handle both linear and nonlinear classification tasks.
- However, when the data is not linearly separable, kernel functions are used to transform the data higher-dimensional space to enable linear separation.
- This application of kernel functions can be known as the “kernel trick”, and the choice of kernel function, such as linear kernels, polynomial kernels, radial basis function (RBF) kernels, or sigmoid kernels, depends on data characteristics and the specific use case

SVM for Classification and Regression

- **SVM for Classification (SVC)**: Uses hyperplanes to separate data into different classes.
- **SVM for Regression (SVR)**: Uses an alternative objective to fit a regression model while minimizing the margin violations.

Linear SVM vs Non-Linear SVM

Linear SVM	**Non-Linear SVM**
It can be easily separated with a linear line.	It cannot be easily separated with a linear line.
Data is classified with the help of hyperplane.	We use Kernels to make non-separable data into separable data.
Data can be easily classified by drawing a straight line.	We map data into high dimensional space to classify.

Advantages of SVM

- Effective in high-dimensional spaces.
- Works well with a clear margin of separation.
- Can handle non-linearly separable data using kernels.
- Robust to overfitting with proper regularization.

Disadvantages of SVM

- Computationally expensive for large datasets.
- Choosing the right kernel and hyperparameters can be complex.
- Less effective with noisy data.

Applications of SVM

- **Text Classification**: Spam detection, sentiment analysis.
- **Image Classification**: Handwritten digit recognition.
- **Medical Diagnosis**: Cancer detection from medical images.
- **Bioinformatics**: Protein classification and gene expression analysis.